Development of a GA4GH Beacon for structured Clinical Data Discovery using the OMOP Common Data Model

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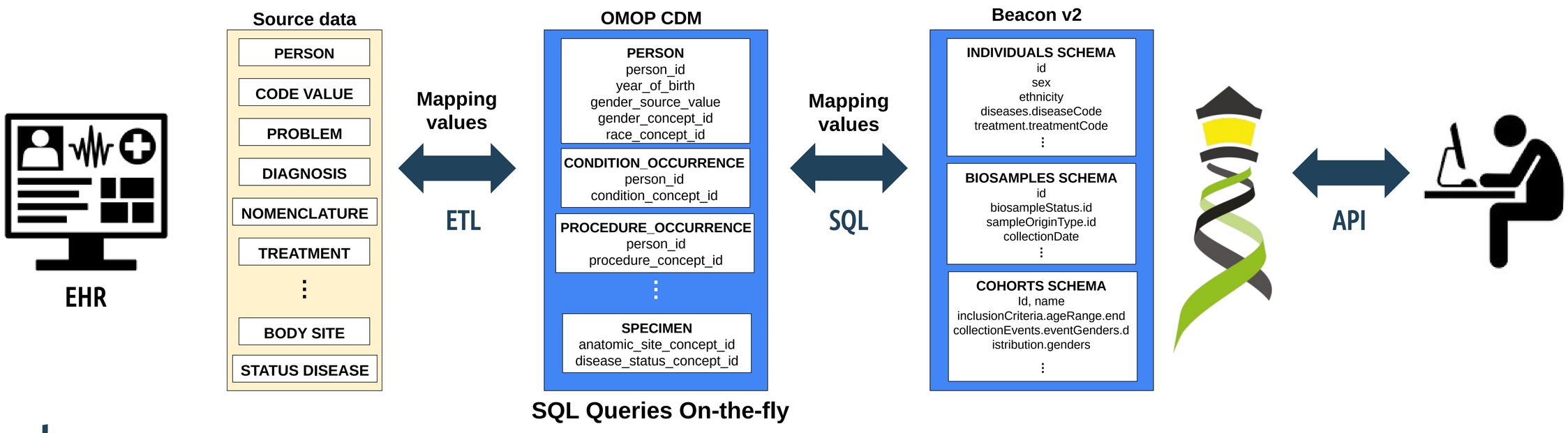
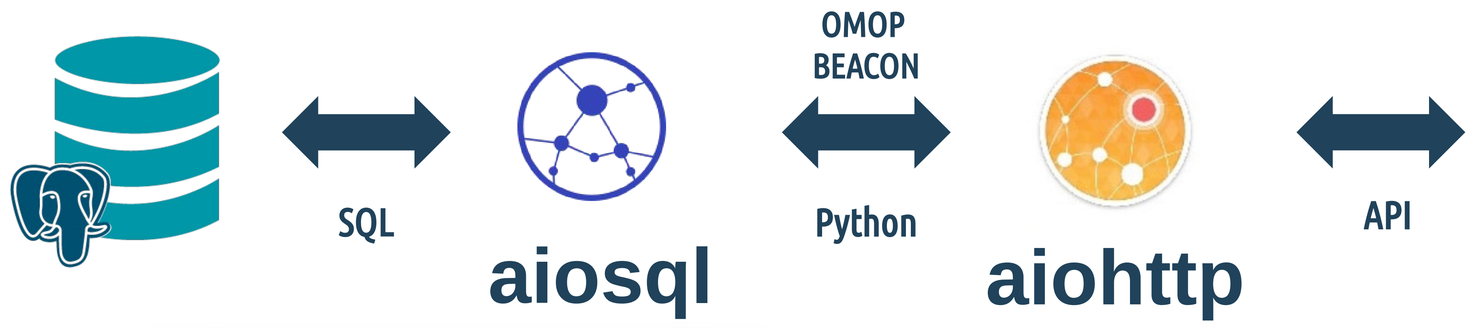
We present the development of **GA4GH Beacon v2** for structured clinical data discovery based on the Observational Medical Outcomes Partnership Common Data Model (OMOP-CDM). Beacon v2 is a standard specification for anonymous **federated data discovery** of genomic and phenoclinic data. Whereas, OMOP is a widely used standardized model for organizing and **harmonizing clinical information** mostly coming from Electronic Health Records (EHRs) data, enabling large-scale data analysis across multiple data sources. This Beacon implementation uses the OMOP-CDM to **enable querying of clinical data sources** for specific phenotypic criteria, such as diagnoses, treatments, and laboratory results. Thus, this application will increase the value of having an OMOP-CDM in the organisations as it will leverage the work of the Beacon implementers and future Beacon Networks (Beacon that query many Beacons).



Methods

- User-friendly interface for querying clinical data sources.
- Address the challenges and opportunities for implementing a GA4GH Beacon for clinical data discovery, including **data privacy and security** considerations, and the potential impact of such a tool for clinical research and healthcare delivery.
- Developed in Python following the EGA-CRG Beacon v2 RI API.




```
select person_id
from cdm.person p
where true and exists (
select 1
from cdm.condition_occurrence co
where p.person_id = co.person_id
and (condition_concept_id = 378001 or condition_concept_id = 4297140)
)
```

```
curl https://beacons.bsc.es/beacon/v2.0.0/individuals/?filters=SNOMED:62106007
```


```
"response": {
  "resultSets": [
    {
      "exists": true,
      "id": "individual",
      "results": [
        {
          "ethnicity": {
            "id": "NCIT:C42331",
            "label": "African"
          },
          "id": "HG00096",
          "info": {
            "eid": "fake1"
          },
          "interventionsOrProcedures": [
            {
              "procedureCode": {
                "id": "OPCS4:L46.3",
                "label": "OPCS(v4-0.0):Ligation of visceral branch of abdominal aorta NEC"
              }
            }
          ],
          "measures": [
            {
              "assayCode": {
                "id": "LOINC:35925-4",
                "label": "BMI"
              },
              "date": "2021-09-24",
              "measurementValue": {
                "unit": {
                  "id": "NCIT:C49671",
                  "label": "Kilogram per Square Meter"
                },
                "value": 26.63838307
              }
            } ...
```


Once a REST request is received, the corresponding model (individuals, cohorts, etc) is invoked with the query parameters, data is then retrieved from the database and converted to the Beacon v2 specification JSON response which is sent back to the client.

Software

- This project is based on the Beacon v2 Python reference implementation and has been developed in the context of the IMPaCT-DATA WP3.5.
- The reference implementation for this development can be downloaded from the following repository: gitlab.bsc.es/impact-data/impd-beacon_omopcdm
- Please follow the instructions provided in the README for installation and usage. The software is released under an Apache 2.0 license.

The main libraries employed are:

- aiohttp for exposing the REST API methods
- aiosql to model the SQL real-time queries against the OMOP database.

References

1. Rambla J. et al. Beacon v2 and Beacon networks: A "lingua franca" for federated data discovery in biomedical genomics, and beyond. Hum. Mutat., 43, 791–799
2. Rueda, M et al. Beacon v2 Reference Implementation: a toolkit to enable federated sharing of genomic and phenotypic data. Bioinformatics, Vol 38, Issue 19, 2022
3. Voss, EA et al. Feasibility and utility of applications of the common data model to multiple, disparate observational health databases. J Am Med Inform Assoc. 22, 2015

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